

if-else Problems

1. Check even or Odd

```
#include <iostream>
#include <string>
using namespace std;
#define g "\033[32m"
#define reset "\033[0m"

// Check whether a number is even or odd.

int main(){
    int num;
    cout <<"Enter a number : ";
    cin >> num;

    string check = (num == 0) ? "Number is zero."
                        : (num % 2 == 0) ? "Even"
                        : (num % 2 != 0) ? "Odd"
                        : "Invalid-input";
    cout <<g << check <<reset <<endl;

    return 0 ;
}
```

2. Check leap Year

```
#include <iostream>
using namespace std;
/* A year is a leap year if:

1.It is divisible by 4, and
2.It is not divisible by 100, unless
3.It is also divisible by 400. */

int main(){
    int year;
    cout <<"Enter Year: ";
    cin >> year;

    if (year % 400 == 0){
        cout<<"leap year";
    }else if (year % 100 == 0){
        cout<<" not a leap year";
    }else if (year % 4 == 0){
        cout <<"leap year";
    }
}
```

```
    }else{
        cout << "not a leap year";
    }
    return 0;
}
```

3. Check number positive negative or Zero

```
#include <iostream>
#include <string>
using namespace std;
#define g "\033[32m"
#define reset "\033[0m"

//Check if a number is positive, negative, or zero.
int main(){
    int n;
    cout <<"Enter a number: ";
    cin >> n;

    string message = (n > 0) ? "Positive"
                        :(n < 0) ? "Negative"
                        :(n == 0) ? "Zero"
                        : "Invalid-input. Nothing";
    cout <<g <<message <<reset <<endl;

    return 0;
}
```

4. Checking Voting Eligibility

```
#include <iostream>
using namespace std;

// Check voting eligibility (age ≥ 18).

int main(){
    int age;
    cout <<"Enter your age: ";
    cin >> age;

    if(age == 0 || age < 0 || age > 100){
        cout <<"Invalid Age."<<endl;
    }else if(age >= 18){
        cout <<"Eligibility for Vote." <<endl;
    }else{
        cout <<"Not Eligibility." <<endl;
    }
}
```

```
    return 0;
}
```

5. Check number divided by 5 and 11

```
#include <iostream>
using namespace std;

int main(){
    //Check whether a number is divisible by 5 and 11.
    int num;
    cout <<"Enter number : ";
    cin >> num;

    if(num == 0){
        cout <<"Number is Zero." <<endl;
    }else{
        if(num % 5 == 0 ){
            cout <<num << "divisible by 5" <<endl;
        }
        if(num % 11 == 0){
            cout<<num <<" divisible by 11"<<endl;
        }
    }if(num % 5 == 0 && num % 11 == 0 && num != 0){
        cout <<num << " is divisible by 5 and 11" <<endl;
    }else{
        cout <<num <<" not divisible by 5 and 11" << endl;
    }
    return 0;
}
```

6. Find the largest in 3 Numbers

```
#include <iostream>
using namespace std;

int main() {
    int n1, n2, n3;

    cout << "Enter n1: ";
    cin >> n1;

    cout << "Enter n2: ";
    cin >> n2;

    cout << "Enter n3: ";
    cin >> n3;
```

```

// Check equality
if (n1 == n2 && n2 == n3) {
    cout << "All numbers are equal" << endl;
}
else {
    if (n1 == n2)
        cout << n1 << " equal to " << n2 << endl;

    if (n2 == n3)
        cout << n2 << " equal to " << n3 << endl;

    if (n1 == n3)
        cout << n1 << " equal to " << n3 << endl;
}

// Find largest
if (n1 > n2 && n1 > n3) {
    cout << "Largest number is " << n1 << endl;
}
else if (n2 > n1 && n2 > n3) {
    cout << "Largest number is " << n2 << endl;
}
else {
    cout << "Largest number is " << n3 << endl;
}

return 0;
}

```

7. Find the largest in two Numbers

```

#include <iostream>
#include <string>
using namespace std;
#define g "\033[32m"
#define reset "\033[0m"

int main(){
    int num_1,num_2;
    cout << "Enter first Number: ";
    cin >> num_1;
    cout << "Enter Second Number: ";
    cin >> num_2;

    string check = (num_1 == num_2) ? "Both are Same"
        : (num_1 > num_2) ? "First is greater than second"
        : "Second is greater than first";
    cout << g << check << reset << endl;
}

```

```
    return 0;
}
```

8. Find lowerCase and UpperCase

```
#include <iostream>
using namespace std;

int main(){
    char ch;
    cout <<"Enter a Character: ";
    cin >>ch;

    if(ch >= 'A' && ch <= 'Z'){
        cout <<"UpperCase.";
    }else{
        cout <<"LowerCase";
    }

    return 0;
}
```

9. Grade the Marks

```
#include <iostream>
using namespace std;

//Grade calculator (A, B, C, D, F).
int main(){
    int marks;
    cout <<"Enter marks: ";
    cin >> marks;

    if(marks < 0 || marks > 100 ){
        cout <<"Invalid input" <<endl;
    }else if(marks == 100){
        cout << "Grade Excellent";
    }else if(marks >= 90 && marks < 100){
        cout<<"Grade A" <<endl;
    }else if(marks >= 80 && marks < 90){
        cout << "Grade B" <<endl;
    }else if(marks >= 70 && marks < 80){
        cout <<"Grade C" <<endl;
    }else if(marks >= 60 && marks < 70){
        cout <<"Grade D" <<endl;
    }else if(marks >= 40 && marks < 60){
        cout <<"Grade D-2" <<endl;
    }else{
        cout << "Grade F(Fail)" <<endl;
    }
}
```

```
    }  
  
    return 0;  
}
```

10. Check whether a character is a vowel or consonant

```
#include <iostream>  
using namespace std;  
  
int main() {  
    char ch;  
  
    cout << "Enter a character: ";  
    cin >> ch;  
  
    if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u' ||  
        ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U') {  
        cout << "Vowel";  
    } else {  
        cout << "Consonant";  
    }  
  
    return 0;  
}
```

11. Check whether a number is a 3-digit number

```
#include <iostream>  
using namespace std;  
  
int main() {  
    int num;  
  
    cout << "Enter a number: ";  
    cin >> num;  
  
    if ((num >= 100 && num <= 999) || (num <= -100 && num >= -999)) {  
        cout << "It is a 3-digit number.";  
    } else {  
        cout << "It is not a 3-digit number.";  
    }  
}
```

```
    return 0;  
}
```